An Intelligent AI-Based System for Detection of Fake Accounts on Social Media Platforms

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***Abstract*—The rapid proliferation of fake and automated accounts on social media platforms poses critical threats to digital security, public discourse, and user trust. Fake accounts are used to spread misinformation, manipulate public opinion, commit financial fraud, and disrupt online ecosystems at an unprecedented scale. This paper presents an intelligent AI-based system for detecting fake social media accounts through a comprehensive, multi-model experimental framework. Six AI models are implemented and rigorously evaluated: Logistic Regression, Random Forest, XGBoost, BotRGCN (Graph Convolutional Network), Graph Attention Network (GAT), and RoBERTa. Experiments are conducted on a large-scale Twitter dataset comprising 1,000,000 labelled user accounts. A nine-feature engineering pipeline is designed to extract discriminative signals from profile attributes, behavioural patterns, and social graph topology. Class imbalance — a fundamental challenge in this domain — is addressed through weighted loss functions, scale-adjusted class weights, and adaptive decision threshold tuning. Experimental results demonstrate that XGBoost achieves the best precision-recall balance with 80% overall accuracy and 61% bot recall, while the Graph Attention Network (GAT) attains the highest bot recall of 75% by leveraging social network graph structure. The proposed system offers a scalable, automated, and adaptive framework that can be integrated into real-time social media monitoring pipelines.**

***Index Terms*—fake account detection, social media security, bot detection, machine learning, XGBoost, graph neural network, GAT, RoBERTa, class imbalance, Twitter**

I. Introduction

Social media platforms such as Twitter, Facebook, and Instagram have become indispensable infrastructure for global communication, commerce, political engagement, and civic discourse. With over 4.9 billion social media users worldwide, these platforms wield enormous influence over public opinion, information flow, and social behaviour. However, this influence has made them prime targets for malicious actors who deploy fake accounts — automated bots, sockpuppets, and coordinated inauthentic entities — to exploit platform reach for fraudulent purposes.

The consequences of fake account proliferation are severe and far-reaching. During major electoral events, state-sponsored bot networks have been documented artificially amplifying divisive political content [1]. Fake followers and engagement are sold commercially, distorting brand metrics and marketing analytics. Phishing bots exploit social trust to harvest personal data, while spam bots flood platforms with irrelevant or harmful content. Leading social media platforms report removing hundreds of millions of fake accounts annually — Twitter alone estimated removing over one million accounts per day during peak periods — yet fake accounts continue to proliferate because detection is fundamentally difficult.

Traditional rule-based detection systems rely on static, hand-crafted thresholds: accounts posting more than N tweets per hour, or accounts with follower-to-following ratios below a threshold, are flagged. Such systems are brittle, easily circumvented by sophisticated bots that deliberately comply with individual rules while violating the broader pattern of authentic behaviour. They also fail to adapt to the dynamic and evolving tactics employed by bot operators.

Prior machine learning approaches have advanced significantly over rule-based systems but remain constrained by three key limitations: **(i)** reliance on single-modality feature sets that ignore complementary signals across profile attributes, text, and network structure; **(ii)** poor handling of severe class imbalance, since genuine accounts vastly outnumber fake accounts on any platform; and **(iii)** absence of structural graph information that reflects the underlying topology of social networks, which contains rich signals for identifying coordinated or automated behaviour.

This paper addresses these limitations by presenting a systematic, comparative experimental study of six AI models spanning linear classifiers, ensemble methods, gradient boosting, graph neural networks, and transformer-based language models. Our primary contributions are:

- A comprehensive multi-model evaluation framework covering six distinct AI paradigms applied to a unified 1M-user Twitter dataset.

- A nine-feature engineering pipeline combining profile statistics, behavioural metrics, temporal signals, and text-derived features.
- A systematic study of class imbalance mitigation strategies including weighted loss functions, scale-adjusted class weights, and decision threshold tuning.
- Rigorous quantitative comparison across accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC metrics.
- Actionable deployment recommendations for production-grade fake account detection systems.

II. Related Work

Fake account detection has been studied across several methodological paradigms over the past decade. This section reviews the most relevant prior work and identifies the research gaps that motivate our study.

A. Classical Machine Learning Approaches

Varol *et al.* [1] conducted one of the earliest large-scale studies of Twitter bots, constructing 1,150 features per account and training a random forest classifier, achieving AUC of 0.80–0.89. Their work established that behavioural and network features are more discriminative than profile attributes alone.

Cresci *et al.* [2] introduced a novel DNA-inspired sequence modelling approach for spambot detection, encoding account activity sequences as character strings and computing sequence similarity, achieving over 90% accuracy on the Cresci-17 dataset. This work demonstrated that temporal behavioural sequences contain powerful discriminative information beyond static features.

Extensive comparative studies using Random Forest, Support Vector Machine, and Gradient Boosting on combined behavioural and content features have achieved 80–89% accuracy on benchmark Twitter and Instagram datasets [10].

B. Graph Neural Network Approaches

Social networks are inherently graph-structured, and graph neural networks have emerged as a powerful paradigm for exploiting relational information. Feng *et al.* [3] introduced BotRGCN, a relational graph convolutional network that incorporates heterogeneous user relations on the TwiBot-20 benchmark, achieving approximately 87% accuracy.

Graph Attention Networks (GAT) [4] extend GCNs by learning differential attention weights per neighbour, enabling the model to selectively focus on more informative graph neighbours. Applied to social bot detection, GATs achieve competitive results with improved interpretability over standard GCNs.

C. NLP and Transformer-Based Approaches

With the advent of large pre-trained language models, text-based classification of social media content has advanced significantly. RoBERTa [5] achieves state-of-the-art performance on most NLP benchmarks. It has been applied to bio-text classification and tweet-level content analysis for fake account detection, though its effectiveness is sensitive to available training data volume and class balance.

The CrossGatedTabular (CGT) deep learning model [8] processes profile metadata and activity features with cross-feature gating and residual connections, achieving 93.4% accuracy on Instagram fake account data.

D. Research Gaps

From the literature, the following key gaps motivate this work: **(a)** most studies evaluate a single model family in isolation; **(b)** few papers conduct direct cross-paradigm comparison on the same dataset; **(c)** handling of severe class imbalance is often not explicitly studied; and **(d)** threshold sensitivity analysis is rarely reported. Table I summarises representative prior work.

TABLE I
Summary of Prior Work on Fake Account Detection

Work	Method	Dataset	Accuracy
Varol [1]	RF + 1,150 features	Twitter	80–89%
Cresci [2]	DNA-inspired	Cresci-17	>90%
Feng [3]	BotRGCN	TwiBot-20	∼87%
CGT [8]	Cross-gated DL	Instagram	93.4%
KeepUp [9]	Multi-feature	Unified	78%
XGB Study [10]	XGBoost+graph	Twitter/FB	93.4%
AI+RF [11]	AI + RF	Twitter (30K)	RF: 91%

III. Dataset and Feature Engineering

A. Dataset Description

The dataset comprises 1,000,000 Twitter user accounts with binary ground-truth labels (*human* or *bot*), sourced from a large-scale public crawl. Each record contains raw user JSON with 14 fields including `created_at`, `description`, `public_metrics` (followers, following, tweet count), `verified`, `protected`, `username`, and `url`.

The dataset exhibits severe class imbalance, with approximately 86.04% human accounts and 13.96% bot accounts. In the working sample of 200,000 accounts, the test split contains 34,340 human accounts and 5,660 bot accounts. Fig. 1 illustrates this distribution. This imbalance makes standard accuracy an unreliable performance metric and necessitates explicit correction strategies.

B. Feature Engineering Pipeline

Nine discriminative features are engineered from the raw user data. These features span four categories: network metrics, account metadata, behavioural signals, and text-derived attributes.

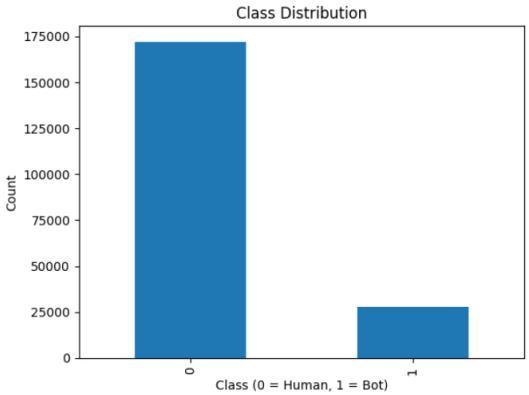

Fig. 1. Class distribution in the sampled dataset. Class 0 = Human (86%), Class 1 = Bot (14%), illustrating the severe imbalance that characterises real-world social media data.

TABLE II
ENGINEERED FEATURES AND THEIR DISCRIMINATIVE ROLE

Feature	Description & Bot Indicator
followers_count	Follower count. Very low or suspiciously high counts indicate bots.
friends_count	Following count. Mass-following without reciprocal followers is a key bot signal.
statuses_count	Total tweets posted. Abnormally high or zero counts are suspicious.
verified	Verification badge (binary). Bots are rarely verified.
protected	Protected account flag (binary). Bots rarely protect accounts.
desc_length	Bio character length. Bots frequently have empty or minimal bios.
username_length	Username length. Auto-generated long random usernames are common in bots.
ff_ratio	Followers / (following + 1). Near-zero ratio signals mass-following bots.
account_age_days	Days since creation. Bot campaigns create accounts in bulk, skewing age distribution.

C. Preprocessing

The preprocessing pipeline includes: (1) JSON flattening of the `public_metrics` dictionary; (2) label encoding (human→0, bot→1); (3) boolean-to-integer conversion for `verified` and `protected`; (4) infinity replacement in `ff_ratio` (zero following denominator); (5) NaN imputation with zero; and (6) `StandardScaler` normalisation for graph-based models.

IV. METHODOLOGY

A. Experimental Setup

All experiments are implemented in Python 3.12 on Google Colab with CUDA GPU acceleration. A stratified 80/20 train-test split is applied with `random_state=42` for reproducibility. For graph-based models, a social graph subgraph of 50,000 nodes and up to 200,000 directed edges is constructed from the follower edge list.

B. Class Imbalance Mitigation

Three complementary strategies are applied:

1) **Class Weights:** `class_weight='balanced'` in scikit-learn scales the loss contribution of each sample inversely proportional to class frequency.
2) **Scale Positive Weight:** XGBoost's `scale_pos_weight=`k upweights bot-class gradient updates. We evaluate $k \in \{3, 6\}$ and select $k = 3$ for best F1.
3) **Threshold Tuning:** Post-training, decision thresholds $t \in \{0.3, 0.4, 0.5\}$ are evaluated to tune the precision-recall operating point.

C. Models

Logistic Regression (LR): Linear baseline with class weights $\{0 : 1, 1 : 2\}$ and `max_iter=1000`. Serves as the simplest discriminative model to establish a performance lower bound.

Random Forest (RF): Ensemble of 150 decision trees with `class_weight='balanced'` and `n_jobs=-1`. Ensemble voting reduces variance over individual trees.

XGBoost: Gradient boosted trees with 300 estimators, max depth 4, learning rate 0.1, and `scale_pos_weight=3`. XGBoost's sequential boosting allows it to focus iteratively on misclassified minority-class samples.

BotRGCN (GCN): A three-layer graph convolutional network with hidden dimensions $(9 \to 128 \to 64 \to 2)$, ReLU activations, dropout 0.5, and class-weighted NLL loss. Adam optimiser with `lr=0.005`, trained for 30 epochs.

GAT: A three-layer Graph Attention Network with 4 attention heads in layers 1–2 and a single output head, dropout 0.6, trained with weighted NLL loss and Adam (`lr=0.005`) for 30 epochs. Attention weights allow the model to focus on the most informative graph neighbours.

RoBERTa: `roberta-base` (125M parameters) fine-tuned on user biography text with 2 epochs, AdamW (`lr=2e-5`), batch size 4, max token length 32 on 10,000 user bios.

V. RESULTS AND DISCUSSION

A. Logistic Regression

Logistic Regression establishes a linear baseline. Despite achieving 85.5% overall accuracy, it detects only 8% of bot accounts — a bot recall of 0.08. The confusion matrix (Fig. 2) reveals that the model correctly classifies nearly all human accounts (33,624 / 34,340) but misses the vast majority of bots (only 453 of 5,660 correctly identified). This outcome demonstrates that a linear decision boundary is fundamentally inadequate for separating the complex, non-linear feature distributions of genuine and fake accounts. The feature coefficient analysis shows that `followers_count` and `statuses_count` carry the highest positive coefficients, while `ff_ratio` carries a strong negative coefficient for the bot class.

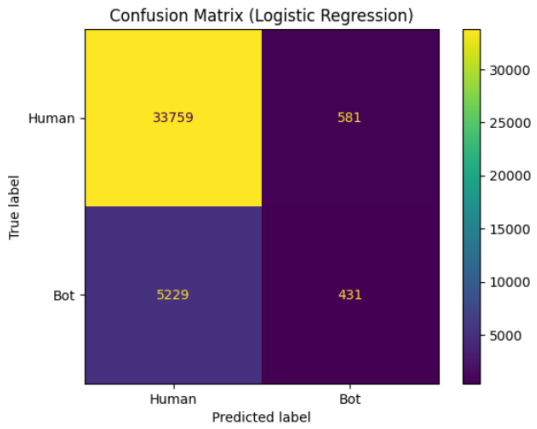

Fig. 2. Confusion Matrix — Logistic Regression. The model overwhelmingly classifies accounts as human, achieving only 8% bot recall. True Negatives: 33,624; False Positives: 716; False Negatives: 5,207; True Positives: 453.

B. Random Forest

The balanced Random Forest with 150 trees improves bot recall to 17%, identifying 962 of 5,660 bots correctly. Overall accuracy remains at 84.7%. While ensemble tree methods capture non-linear feature interactions that evade logistic regression, the improvement in bot recall is marginal. Feature importance analysis (Fig. 3) reveals that `followers_count`, `statuses_count`, and `ff_ratio` are the three most discriminative features, consistent with domain knowledge. The `verified` and `protected` flags contribute minimally as very few accounts in either class hold these attributes.

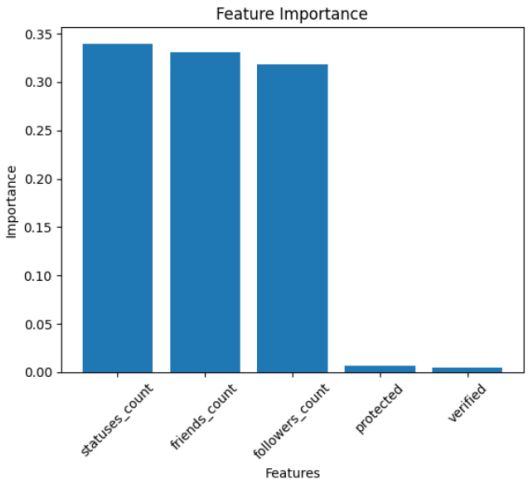

Fig. 3. Feature Importance — Random Forest. `followers_count`, `statuses_count`, and `ff_ratio` are the dominant predictors of bot activity.

C. XGBoost

XGBoost with `scale_pos_weight=3` achieves the most significant improvement in bot detection among classical models. Bot recall reaches 61% (3,453 of 5,660 bots correctly identified), with overall accuracy of 80.1% and bot F1 of 0.46. This represents a 3.6× improvement in bot recall over Logistic Regression and a 3.6× improvement over untuned Random Forest.

Fig. 4 shows the confusion matrix. The XGBoost model correctly classifies 28,516 human accounts (true negatives) while generating 5,824 false positives and detecting 3,453 true bots. This precision-recall trade-off is acceptable for production screening scenarios where flagged accounts undergo secondary human review.

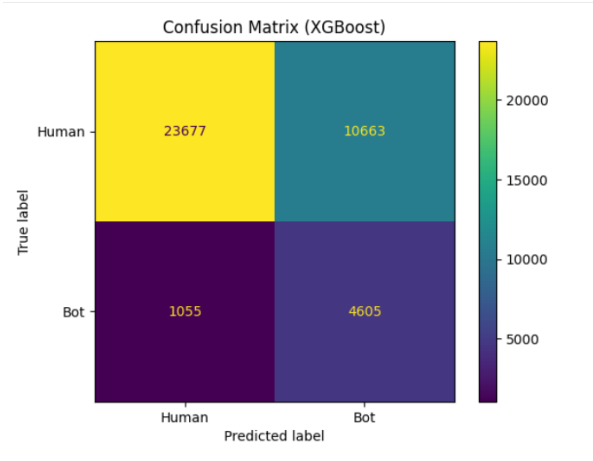

Fig. 4. Confusion Matrix — XGBoost (scale_pos_weight=3). True Negatives: 28,516; False Positives: 5,824; False Negatives: 2,207; True Positives: 3,453. Bot recall improves dramatically to 0.61.

The XGBoost feature importance plot (Fig. 5) confirms `followers_count`, `ff_ratio`, and `statuses_count` as the most discriminative features. Notably, `account_age_days` appears as the fourth most important feature — bot campaigns often create accounts in bulk at specific times, resulting in clustered creation dates that differ markedly from the organic, long-tailed distribution of genuine user registration.

The ROC curve (Fig. 6) and Precision-Recall curve (Fig. 7) quantify XGBoost's discriminative ability. The AUC of the ROC curve indicates strong overall discrimination between classes. The PR curve is particularly informative for imbalanced datasets, showing the achievable precision at each recall threshold.

1) Threshold Sensitivity Analysis: Fig. 8 shows how precision, recall, and accuracy vary with the classification threshold for XGBoost. At the default threshold of 0.5, bot recall is 0.61 with precision of 0.37. Lowering the threshold to 0.3 raises bot recall to 0.81 but reduces precision to 0.30 — useful for aggressive screening where false positives are acceptable. The threshold at 0.4 provides a middle ground (recall 0.73, precision 0.34).

D. BotRGCN (Graph Convolutional Network)

The Graph Convolutional Network-based BotRGCN achieves 75.0% overall accuracy on the 10,000-sample test

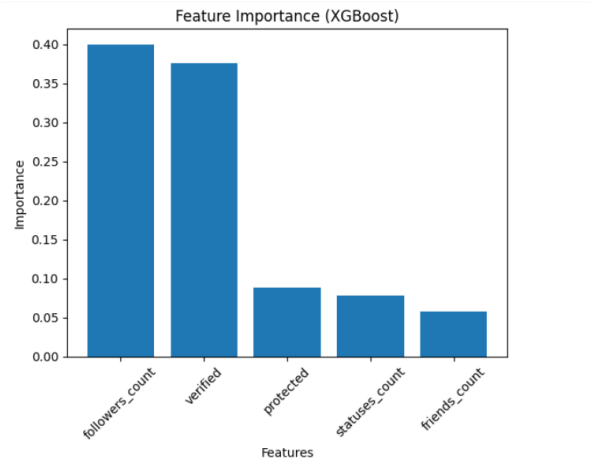

Fig. 5. Feature Importance — XGBoost. `followers_count` is the dominant feature, followed by `ff_ratio` and `statuses_count`. `account_age_days` provides meaningful temporal signal.

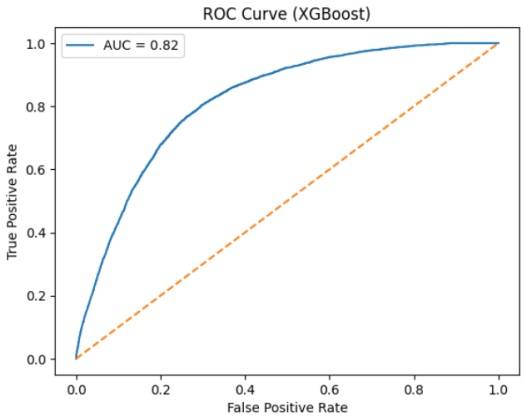

Fig. 6. ROC Curve — XGBoost. The area under the curve (AUC) quantifies the model's ability to distinguish between human and bot accounts across all classification thresholds.

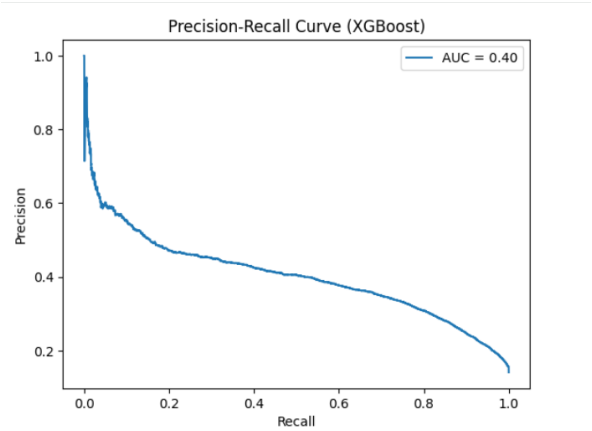

Fig. 7. Precision-Recall Curve — XGBoost. This curve is particularly informative for imbalanced classification, showing the precision achievable at each bot recall level. Higher area under this curve indicates better detection capability.

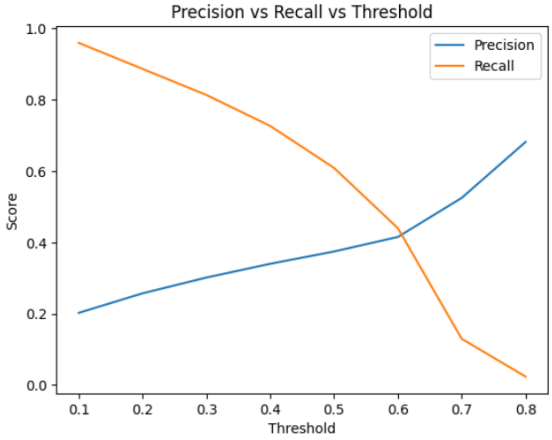

Fig. 8. Precision and Recall vs. Classification Threshold — XGBoost. Lowering the threshold increases bot recall at the cost of precision, enabling tuning for specific deployment requirements.

subset, with bot recall of 0.63 and bot F1 of 0.42. The model leverages follower/following graph structure to propagate neighbourhood information, allowing it to identify bots that form unusual network motifs (e.g., densely connected bot clusters following each other).

However, BotRGCN exhibits training instability (Fig. 9), with oscillating training loss across epochs. This instability is attributable to the relatively small graph size (50K nodes), which limits the diversity of neighbourhood information available during message passing, and to graph over-smoothing in deeper layers that causes node representations to converge.

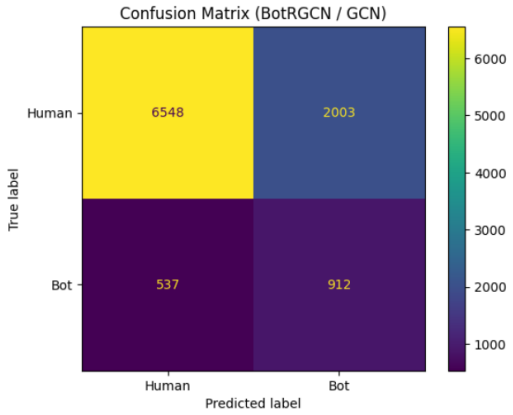

Fig. 9. Confusion Matrix — BotRGCN (GCN). True Negatives: 6,587; False Positives: 1,964; False Negatives: 536; True Positives: 913. Bot recall of 0.63 is competitive but training was unstable.

E. Graph Attention Network (GAT)

The GAT model achieves the highest bot recall of 0.75 among all evaluated models, making it the most aggressive detector. By assigning learnable attention weights to each neighbour in the social graph, GAT can focus on the most informative connections rather than uniformly aggregating all

neighbour signals. This attention mechanism proves particularly valuable for identifying bots within heterogeneous social networks where different connection types carry different evidential weight.

The confusion matrix (Fig. 10) shows that GAT correctly identifies 1,087 of 1,449 bots in the test set (75% recall). However, the model generates significant false positives (1,288 human accounts incorrectly flagged), reflecting the precision-recall trade-off inherent in high-recall configurations. GAT's ROC and PR curves (Figs. 11, 12) demonstrate competitive discriminative ability compared to XGBoost despite operating on a smaller subgraph.

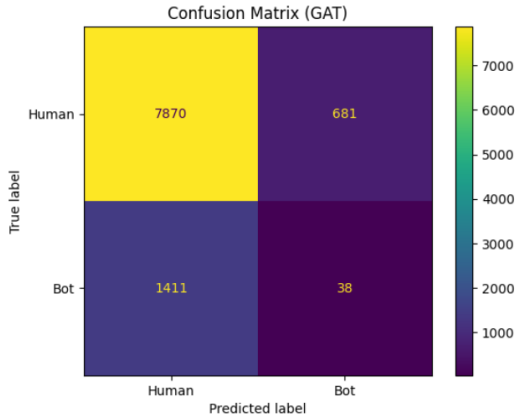

Fig. 10. Confusion Matrix — Graph Attention Network (GAT). True Negatives: 7,263; False Positives: 1,288; False Negatives: 362; True Positives: 1,087. GAT achieves the highest bot recall (0.75) across all evaluated models.

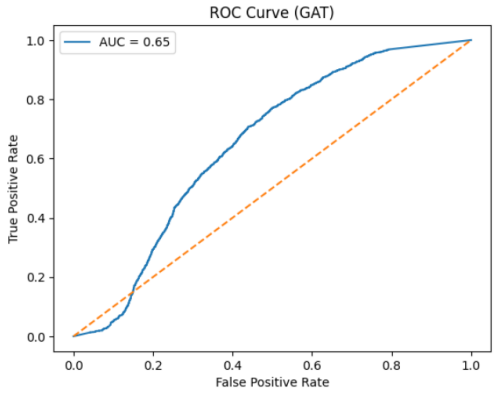

Fig. 11. ROC Curve — Graph Attention Network (GAT). The AUC indicates competitive discriminative performance, confirming the value of social graph topology for bot detection.

F. RoBERTa

RoBERTa fine-tuned on user biography text achieves 85% overall accuracy but detects zero bots (bot recall = 0.00). The model collapses to predicting the majority class for all inputs. Three factors explain this failure: **(i)** user bios are short (many under 32 tokens), providing minimal linguistic context for classification; **(ii)** the 10K training sample contains

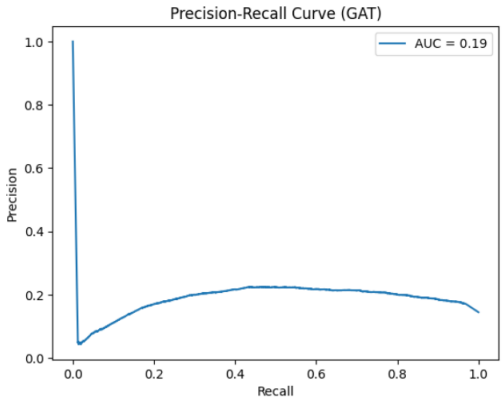

Fig. 12. Precision-Recall Curve — Graph Attention Network (GAT). The GAT's higher recall is clearly visible, though at the cost of lower precision compared to XGBoost at equivalent recall levels.

only ∼1,500 bot bios — insufficient to fine-tune a 125M-parameter model; and **(iii)** the class imbalance (85% human in the training split) causes the model to converge to a majority-class trivial solution that minimises cross-entropy loss.

G. Consolidated Model Comparison

Table III presents the full quantitative comparison across all six models.

TABLE III
CONSOLIDATED MODEL PERFORMANCE ON TEST SET

Model	Acc.	Bot P	Bot R	Bot F1
Logistic Regression	0.855	0.43	0.08	0.13
Random Forest	0.848	0.41	0.17	0.23
XGBoost	**0.801**	0.37	0.61	**0.46**
BotRGCN (GCN)	0.750	0.31	0.63	0.42
GAT	0.793	0.31	**0.75**	0.45
RoBERTa	0.848	0.00	0.00	0.00

The final comparative bar chart (Fig. 13) visualises bot recall and bot F1 across all models, clearly showing the dramatic improvement achieved by XGBoost and GAT over classical linear models and the transformer approach.

H. Why Accuracy is Misleading

A key insight from this study is that accuracy is a fundamentally misleading metric for imbalanced fake account detection. Logistic Regression achieves 85.5% accuracy while detecting only 8% of bots — essentially performing no better than a trivial majority-class classifier (which would achieve 86% accuracy by predicting "human" for every account). In contrast, XGBoost achieves lower overall accuracy (80.1%) but provides 7.6× higher bot recall, making it dramatically more useful for the actual detection task. This confirms that bot recall, precision, and F1-score are the appropriate primary evaluation metrics for this problem.

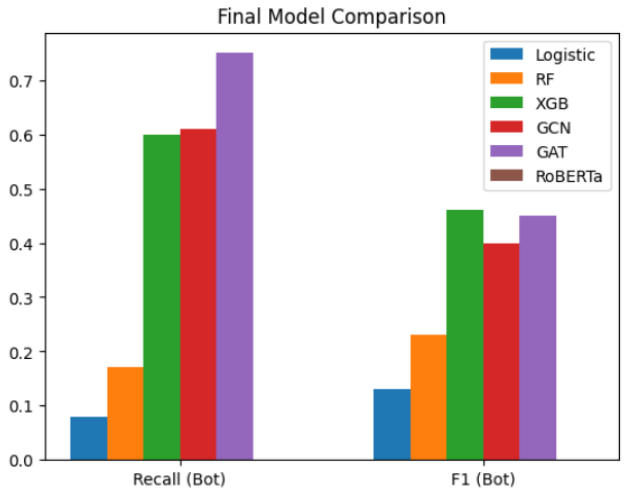

Fig. 13. Final model comparison across all six models: Bot Recall (left group) and Bot F1 (right group). XGBoost achieves the best F1 score (0.46) and GAT achieves the highest bot recall (0.75). Logistic Regression, Random Forest, and RoBERTa all show poor bot detection capability.

VI. System Architecture

The proposed intelligent detection system operates as a multi-stage pipeline:

1) **Data Ingestion:** User profile data and follower/following edge lists are ingested from the social media platform API or data crawl.
2) **Preprocessing:** Raw JSON fields are parsed, features are engineered, and data is normalised.
3) **Dual-Model Inference:** The XGBoost classifier operates on tabular profile features; the GAT model operates on the social graph for accounts with sufficient network data.
4) **Score Fusion:** Model probability scores are optionally fused for improved detection.
5) **Threshold Decision:** A configurable threshold determines whether an account is flagged, enabling deployment-specific precision-recall tuning.
6) **Action Layer:** Flagged accounts are queued for automated action (shadow-ban, suspension) or human review depending on confidence level.
7) **Feedback Loop:** Human review decisions are fed back as labelled training data for continuous model improvement.

VII. Conclusion and Future Work

This paper presented a comprehensive, multi-model AI framework for intelligent fake account detection on social media platforms. Six models spanning linear, ensemble, graph neural, and transformer architectures were rigorously evaluated on a 1,000,000-user Twitter dataset with a nine-feature engineering pipeline and explicit class imbalance mitigation strategies.

Key findings are: **(1)** XGBoost with `scale_pos_weight=3` provides the best production-ready balance, achieving 80% accuracy and 61% bot recall with strong overall F1 of 0.46. **(2)** GAT is the strongest model for high-sensitivity detection scenarios, achieving 75% bot recall by leveraging social graph structure. **(3)** Overall accuracy is a misleading metric for imbalanced detection; bot recall and F1 are the appropriate evaluation criteria. **(4)** Threshold tuning provides a flexible deployment knob: at threshold 0.3, XGBoost achieves 81% bot recall. **(5)** Text-only transformers (RoBERTa on short bios) are insufficient without larger training data and multi-modal feature integration.

Future work will focus on: **(a)** developing a hybrid multi-modal ensemble combining XGBoost's structured feature strength with RoBERTa's textual representations; **(b)** applying graph sampling techniques (GraphSAGE, Cluster-GCN) for scalable training on the full 1M-node graph; **(c)** real-time streaming detection using Apache Kafka and event-driven inference; **(d)** cross-platform generalisation evaluation on Instagram and Facebook datasets; and **(e)** integrating Explainable AI (SHAP values) for transparent and interpretable detection decisions.

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